

MAEG4060 Course Project

User Guide

Interactive Physical Simulation of Off-Road Vehicle Dynamics

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1 How to start the application

- Open the `Game_Build` folder in your submission package (or run from the repository root after a local build; see below).
- **Double-click** `maeg4060_stage2.exe` in that folder to launch the program.
- Keep the executable beside the `Assets` folder so paths such as `Assets/models/model_jeep.x` resolve. If models fail to load, confirm the working directory contains `Assets`.

1.1 Building from source (optional)

The project is a CMake target (`maeg4060_stage2`) that needs legacy **D3DX** headers and import libraries (`d3dx9.h`, `d3dx9math.h`, etc.), which are *not* supplied by the Windows 10/11 SDK alone. CMake resolves them in one of two ways: (1) set `DXSDK_DIR` to the root of the **June 2010 DirectX SDK** (must contain `Include` and `Lib`); or (2) leave `DXSDK_DIR` unset and let CMake download the `Microsoft.DXSDK.D3DX` NuGet package into the build directory on first configure (requires network access; configure may take several minutes).

Recommended (matches `scripts/configure_build_release.bat`): run `vcvars64.bat` from your Visual Studio installation so `cl` is on `PATH`, then from the repository root use the Ninja generator and select `MSVC` explicitly, for example:

```
cmake -B build -G Ninja -DCMAKE_BUILD_TYPE=Release ^
      -DCMAKE_CXX_COMPILER=cl -DCMAKE_C_COMPILER=cl
cmake --build build
```

On some setups the Visual Studio CMake generator fails to detect a C++ compiler even after `vcvars64`; Ninja with `-DCMAKE_CXX_COMPILER=cl` avoids that. If you change the generator in an existing build tree, delete the `build` folder (or remove `CMakeCache.txt` and the `CMakeFiles` directory) before re-running `cmake`.

If the compiler stops with **C1060** (out of heap space), retry the build with less parallelism, e.g. `cmake -build build -j 1`, and close other heavy applications.

Alternatively, from the repository root: `cmake -B build -G "Visual Studio 17 2022" -A x64`, then `cmake -build build -config Release`, provided the toolset is detected correctly.

To package binaries: run `scripts/package_game_build.bat` to populate `Game_Build`, or `cmake -install build -prefix Game_Build`.

1.2 GitHub Actions (optional)

If `.github/workflows/windows-build.yml` is enabled on your fork, the **Windows build** workflow on `windows-latest` produces an artifact (zip) containing `maeg4060_stage2.exe`,

D3DX9_43.dll when the NuGet D3DX path is used, and a copy of **Assets/**. Download the artifact, unzip on Windows, and run the executable from that folder (keep **Assets** beside the .exe).

2 Hardware and software requirements

- **Platform:** Windows PC (64-bit Windows recommended).
- **Graphics API:** The application uses **DirectX 9.0c**. You need a DirectX 9-compatible GPU and the **DirectX 9 End-User Runtime** (or equivalent system components) installed as required by your Windows version.
- **Display:** A desktop resolution of at least **1280×720** is recommended; other resolutions are supported when the window is resized.
- **Optional input:** An **XInput-compatible gamepad** (e.g. Xbox controller) for analog control. Keyboard control works without a gamepad.

3 Controls

3.1 Keyboard (WASD and related keys)

Input is handled via the Win32 keyboard API (`GetAsyncKeyState`) with a vehicle-style layout, for example:

Input	Action
W / S	Forward / reverse (throttle)
A / D	Steer left / right
Space	Brake (or hand brake, depending on build)
M	Pause / resume windmill motion (Maya animation when present, or simple spin for the static mesh f
Esc	Quit application

Exact key bindings may match the on-screen HUD or readme in the binary package; the layout above reflects the project design (WASD movement with separate brake).

3.2 Gamepad (XInput)

With an XInput controller connected, the application maps analog axes and triggers for smoother control than the keyboard:

Control	Action
Left stick	Steering
Right trigger	Throttle (accelerate)
Left trigger	Brake / partial reverse

Triggers provide analog throttle similar to a real vehicle; keyboard W/S act as digital throttle inputs.

4 Troubleshooting

- **Missing DLLs or DirectX errors:** Install or repair the DirectX 9 end-user runtime and ensure GPU drivers are up to date.
- **Black screen or failed device creation:** Try windowed mode, update graphics drivers, and confirm no other application has exclusive full-screen access to the GPU.
- **Assets not found:** Run from `Game_Build` with the supplied `Assets` (or equivalent) folder beside the executable. For static prop replacements, place OBJ/MTL files under `Assets/models/replacements` (see `Assets/models/replacements/README.md`).
- **Gamepad not detected:** Use a wired connection or official wireless adapter; confirm the controller appears in Windows as an XInput device.